



Written exam for MCE2 and PED2 students in
Control of Electrical Drive Systems and Converters
Tuesday 18 June 2026 at 9:30 (4 hours + 15 minutes)

- Remember to write your study number on all answer sheets. Do not write your name as your evaluation is anonymous.
- **All kinds of communication with other students or the outside world are strictly forbidden. Use of ChatGPT and other AI tools is strictly forbidden.**
- All aids are permitted, including textbooks, lecture slides, personal notes, exercise solutions, advanced calculators, and computers. Use of the internet is **not** permitted. Any materials from Moodle must be downloaded to the computer in advance, and OneDrive must be set to offline mode. Use of MATLAB/Simulink is allowed; however, use of PLECS is prohibited. Copying and pasting figures or equations from software, textbooks, or other sources is not allowed.
- Questions to the exam questions must always be addressed to the invigilators.
- Tablet or mobile phone may be brought to the exam only for use in connection with uploading handwritten pages. You may choose to make handwritten answers or to answer on your computer or a combination. You can scan your exam paper via your Iphone or IPad. Iphone/IPad must be hidden during the exam. It is your responsibility to ensure high quality of the uploaded pictures/document.
- Upload your answer sheets as a PDF file, preferably as 1 common file.
- Your answers must be uploaded to Digital Exam during the last quarter from 13:30-13:45. After 13:30 you are no longer allowed to answer any of the assignments. Invigilation is to be contacted in case you want to submit early (but not during the last 30 minutes).
- It is not allowed to access recorded lectures during the examination.
- This examination set contains **4** problems.
- You are welcome to take the exam papers/questions with you after the exam has finished. However, you are NOT allowed to take them with you if you leave before the exam has ended.



Part 1 Converters (50 %)

Problem 1 (30 %)

- 1) Consider a full-bridge single-phase VSI with bipolar modulation. Draw the schematics, switching states table, waveform for carrier and references, gate signals for S+ and S- and the output voltage V_{an} , V_{bn} and V_{ab} considering sinusoidal PWM. What is the maximum output voltage RMS for the fundamental with respect to input dc voltage. What is voltage THD, its formula and how and why it depends on switching frequency. What will be the impact on THD if unipolar PWM will be used (15%)
- 2) Space Vector Modulation: Define it and explain why in vector control of ac motor SVM it is preferred against PWM sinusoidal. Draw the space vector representation showing all active vectors in all 6 sectors and the on-time periods T_1 , T_2 and T_0 . What is the maximum output voltage fundamental RMS as a ratio of the input dc voltage with demonstration.? (15%)

Problem 2 (20 %)

- 1) A three-phase motor-drive inverter uses an IGBT module connected to a DC-link voltage of:

$$V_{dc} = 650 \text{ V}$$

For one IGBT, assume the following operating conditions and datasheet parameters:

Category	Parameter	Value
Current	Average current	$I_{avg} = 8 \text{ A}$
Current	RMS current	$I_{rms} = 10 \text{ A}$
Conduction characteristic	Forward voltage drop	$V_F = 1.5 \text{ V}$
Conduction characteristic	Dynamic on-resistance	$r_{on} = 5 \text{ m}\Omega$
Switching energy	Turn-on energy	$E_{on} = 0.9 \text{ mJ}$
Switching energy	Turn-off energy	$E_{off} = 0.6 \text{ mJ}$
Datasheet test condition	Test voltage	$V_t = 600 \text{ V}$
Datasheet test condition	Test current	$I_t = 15 \text{ A}$
Operating condition	Switching frequency	$f_{sw} = 10 \text{ kHz}$
Thermal data	Ambient temperature	$T_{amb} = 45 \text{ }^\circ\text{C}$
Thermal data	Maximum junction temperature	$T_{j,max} = 125 \text{ }^\circ\text{C}$
Thermal data	Junction-to-case thermal resistance	$R_{jc} = 0.7 \text{ }^\circ\text{C/W}$
Thermal data	Case-to-heatsink thermal resistance	$R_{cs} = 0.2 \text{ }^\circ\text{C/W}$

Answer the following questions:

- a. Explain why an IGBT may be preferred over a MOSFET in a high-voltage motor-drive inverter.
 - b. Compute the IGBT conduction loss.
 - c. Compute the switching loss.
 - d. Compute the total loss of the IGBT.
 - e. Determine the maximum allowable heatsink-to-ambient thermal resistance, R_{sa} , required to keep the junction temperature below $T_{j,max}$.
 - f. If the switching frequency is doubled, explain what happens to the switching loss and the required heatsink size.
- 2)
 - a. Explain why a simple diode bridge rectifier with a DC-link capacitor gives poor input-current quality.
 - b. Passive PFC and active boost PFC are two methods used to improve the input current. Explain the difference between them.

Part 2 Electrical Drives (50 %)

Problem 3 (25%)

A permanent magnet synchronous machine has the following data:

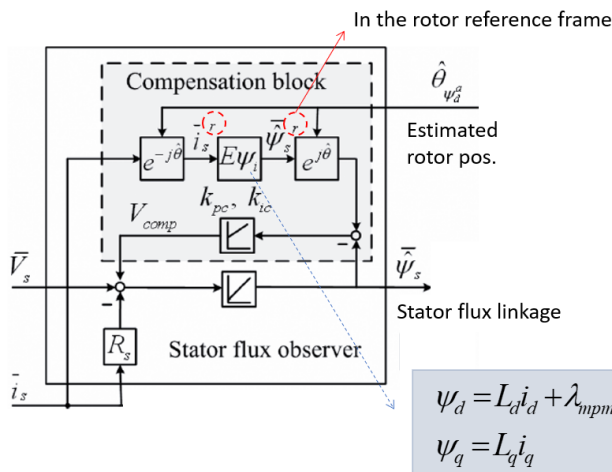
DATA OF THE 2.7-KW EIGHT-POLE SPMSM

Rated values		Parameters	
Current (RMS)	7.4 A	Stator resistance R_s	1.2 Ω
Torque	5.8 N·m	$d(q)$ -axis inductance L	5.5 mH
Frequency	300 Hz	Flux linkage by PM λ_m	0.1213 Wb-t
Speed	4500 r/min	Total inertia J	0.0125 kg·m ²

In field-oriented control, we will normally introduce back-EMF decoupling to the d- and q-axis current controllers.

- In steady state, if the back-EMF decoupling is removed, will there be any steady state error in tracking the current reference?
- Suppose the motor operates at 1200 rpm stably. The load torque is the rated torque. In this steady state, please calculate the outputs of the PI controllers in the two current loops for regulating the d- and q-axis currents when 1) there is no back-EMF decoupling; 2) back-EMF decoupling is applied. (You do not need to know the PI parameters.)

The following is the closed-loop flux observer used for estimating the rotor position.



- In the above implementation, the dc-drift compensation term (V_{comp}) is introduced before the integrator (i.e., the stator flux linkage $= \int (\bar{V}_s - R_s \bar{i}_s + \bar{V}_{comp}) dt$). Can we move this compensation term after the integrator (i.e., $\int (\bar{V}_s - R_s \bar{i}_s) dt + \bar{V}_{comp}$)? Please give your thoughts.

The motor is under field-oriented control, and it operates at 3000 rpm with an encoder measuring the position and speed. The load torque is proportional to the speed. At a particular moment when the rotor d-axis is aligned with the phase-a axis, the encoder suddenly loses its power supply,

- Please explain what happens to the motor based on your understanding of the field-oriented controller.

Problem 4 Induction machine (25%)

The mathematical model of an induction machine may be expressed as:

The stator side:

The rotor side:

$$\begin{aligned} u_{qs} &= r_s i_{qs} + p \lambda_{qs} + \omega_e \lambda_{ds} & \lambda_{qs} &= L_s i_{qs} + L_m (i_{qs} + i_{qr}) & u_{qr} &= r_r i_{qr} + p \lambda_{qr} + (\omega_e - \omega_r) \lambda_{dr} & \lambda_{qr} &= L_r i_{qr} + L_m (i_{qs} + i_{qr}) \\ u_{ds} &= r_s i_{ds} + p \lambda_{ds} - \omega_e \lambda_{qs} & \lambda_{ds} &= L_s i_{ds} + L_m (i_{ds} + i_{dr}) & u_{dr} &= r_r i_{dr} + p \lambda_{dr} - (\omega_e - \omega_r) \lambda_{qr} & \lambda_{dr} &= L_r i_{dr} + L_m (i_{ds} + i_{dr}) \end{aligned}$$

where λ represents the flux linkage, and $p = \frac{d}{dt}$. The voltage supplied to the rotor windings is zero, assuming it is a squirrel cage induction machine, i.e., $u_{qr} = u_{dr} = 0$.

Now we will perform a flux-oriented controller. The chosen flux vector is:

$$\bar{\lambda}_{dqm} = \lambda_{dm} + j \lambda_{qm} = [L_m (i_{ds} + i_{dr})] + j [L_m (i_{qs} + i_{qr})] = L_m \bar{i}_{dqs} + L_m \bar{i}_{dqr} \quad (1)$$

The d-axis of the chosen rotating dq-frame will always be aligned with $\bar{\lambda}_{dqm}$.

The machine equations can be re-arranged in the following vector form as:

Stator side voltage and flux linkage equations:

$$\bar{u}_{dqs} = r_s \cdot \bar{i}_{dqs} + p \bar{\lambda}_{dqs} + j \omega_e \bar{\lambda}_{dqs} = r_s \cdot \bar{i}_{dqs} + (p + j \omega_e) \bar{\lambda}_{dqs} \quad \bar{\lambda}_{dqs} = L_s \bar{i}_{dqs} + L_m \bar{i}_{dqr}$$

Rotor side voltage and flux linkage equations:

$$0 = r_r \cdot \bar{i}_{dqr} + p \bar{\lambda}_{dqr} + j (\omega_e - \omega_r) \bar{\lambda}_{dqr} = r_r \cdot \bar{i}_{dqr} + (p + j s \omega_e) \bar{\lambda}_{dqr} \quad \bar{\lambda}_{dqr} = L_r \bar{i}_{dqr} + L_m \bar{i}_{dqs}$$

- From the defined flux linkage in (1), re-express the rotor current $\bar{i}_{dqr}$ as a function of variables $\bar{\lambda}_{dqm}$ and $\bar{i}_{dqs}$. Substitute this rotor current into the rotor flux linkage equation to express the rotor flux linkage as a function of variables $\bar{\lambda}_{dqm}$ and $\bar{i}_{dqs}$.
- Substitute the reformed rotor current and rotor flux linkage expressions into the rotor voltage equation to obtain an equation that involves variables $\bar{\lambda}_{dqm}$ and $\bar{i}_{dqs}$ only (i.e., get rid of the rotor current and rotor flux linkage).
- Decompose the above vector equation into its d- and q-components ($d = \text{real}()$ and $q = \text{imag}()$). Apply the condition that the d-axis is always aligned with the flux vector $\bar{\lambda}_{dqm}$ to simplify the equations.
- Use the q-axis equation to determine how the slip speed $s \omega_e$ is calculated. Use the d-axis equation to determine how the i_{ds} current should be determined.
- Based on the above two equations, draw a controller topology where the inputs are $|\bar{\lambda}_{dqm}| = \lambda_{dqm}$ and the torque reference (which is determined by the speed loop's controller). The outputs are i_{ds} and i_{qs} command values. Assuming the torque $T_e = K i_{qs}$, where K is the torque constant.